Discrete Structures: The Foundations – Logic and Proofs

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Outline

- Propositional Logic
- Applications of Propositional Logic
- Propositional Equivalences
- Predicates and Quantifiers
- Nested Quantifiers
- Rules of Inference
- Introduction to Proofs

Compound Propositions

- Common logical connectives
  - Negation: $\neg$. E.g. $\neg p$
  - Conjunction: $\wedge$. E.g. $p \wedge q$
  - Disjunction: $\vee$. E.g. $p \vee q$
  - Exclusive or: $\oplus$. E.g. $p \oplus q$
  - Conditional (implication): $\rightarrow$. E.g. $p \rightarrow q$
  - Biconditional: $\leftrightarrow$. E.g. $p \leftrightarrow q$

  p *and* q are propositional variables

Conditional Statement 1

- Let p and q be propositions. The **conditional statement** $\boldsymbol{p \rightarrow q}$ is the proposition "**if $\boldsymbol{p}$, then $\boldsymbol{q}$**".
- The conditional statement $p \rightarrow q$ is false when p is true and q is false, and true otherwise.
  - In the conditional statement $p \rightarrow q$, $\boldsymbol{p}$ **is called the *hypothesis*** (or *antecedent* or ***premise***) and $\boldsymbol{q}$ **is called the *conclusion*** (or *consequence*).
- Truth table for conditional statement

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	**T**	**T**
F	**F**	**T**

Conditional Statement 2

- A conditional statement is also called an **implication.**
- Note that the statement $p \rightarrow q$ is true when both p and q are true and when p is false (**no matter what truth value q has**)
  - Think about "If I am elected, then I will lower taxes"
    - When has this politician broken the campaign pledge ?
      - Only when the politician is elected but does not lower taxes
    - In other cases, we can't say it's wrong, i.e., it's not wrong (true)
  - Think about "Evidences of criminal behaviors imply the suspect would be guilty."
    - Without evidence, we can only tell the suspect is innocent.

Understand Implication 1

- Conditional statements are widely used, and there is a variety of terminology used:

"if p, then q"	"p implies q"
"if p, q"	"***p* only if *q***"
"p is **sufficient** for q"	"a sufficient condition for q is p"
"q if p"	"q whenever p"
"q when p"	"q is **necessary** for p"
"a necessary condition for p is q"	"q follows from p"
"q **unless** $\neg p$"	

  - "***p* only if *q***" means "q is **necessary** for p"
  - "q **unless** $\neg p$" means if $\neg p$ is false, then q must be true

Understand Implication 2

- Same idea with different ways: p is $x = 2$, q is $x^2 - 5x + 6 = 0$, and we want to say $p \to q$
  - "If $x = 2$, then $x^2 - 5x + 6 = 0$."
    - Can we say "If $x^2 - 5x + 6 = 0$, then $x = 2$"? No: try $x = 3$.
  - "$x^2 - 5x + 6 = 0$ when/whenever $x = 2$."
  - "$x = 2$ is sufficient for $x^2 - 5x + 6 = 0$."
  - "$x^2 - 5x + 6 = 0$ is necessary for $x = 2$."
  - "$x = 2$ only if $x^2 - 5x + 6 = 0$."
  - "A necessary condition for $x = 2$ is $x^2 - 5x + 6 = 0$."
    - They all show $x^2 - 5x + 6 = 0$ is necessary
    - This means "without $x^2 - 5x + 6 = 0$ we can't have $x = 2$".
    - "If $x^2 - 5x + 6 \neq 0$, then $x \neq 2$."
      - Why it's the same idea? See Contrapositive in next section.

Understand Implication 3

- Example: "If the moon is made of green cheese, then I have more money than Bill Gates."
  - Solution: the premise is false, so the statement is true. (Take another look at the truth table!)

- Example: "If 1+1=3, then 3+ 4=5."
  - Solution: the premise is false, so the statement is true. (Take another look at the truth table!)

- Given a truth table, can you give the logic expression?

Contrapositive, Converse, Inverse $_1$

- $p \rightarrow q$ can be translated as "q is **necessary** for p"
  - We want to understand why in $p \rightarrow q$, q is **necessary**
    - Without q we can't have p. If q is false, p can't be true.
    - There are only two values for q, so q must be true. This is "indirectly" implied by p.
  - This means $p \rightarrow q$ equals to $\neg q \rightarrow \neg p$
    - They always have the same truth values!
  - $\neg q \rightarrow \neg p$ is the **contrapositive** of $p \rightarrow q$
- Conditional promise "If I am elected, then I will lower taxes" – what if the condition is false?
  - We need to test $\neg p \rightarrow \neg q$ ---- **inverse** of $p \rightarrow q$
- Similarly we have **converse** of $p \rightarrow q$: $q \rightarrow p$

Contrapositive, Converse, Inverse 2

- **Example**: What are the contrapositive, the converse, and the inverse of the conditional statement "If it is raining, then the home team wins."
  - **Solution**:

contrapositive	$\neg \boldsymbol{q} \rightarrow \neg \boldsymbol{p}$	**"If the home team does not win, then it is not raining."**
converse	$q \rightarrow p$	"If the home team wins, then it is raining."
inverse	$\neg p \rightarrow \neg q$	"If it is not raining, then the home team does not win."

Contrapositive, Converse, Inverse 3

- $p \rightarrow q$ can be translated as "q is **necessary** for p"
  - This means $p \rightarrow q$ is equivalent to $\neg q \rightarrow \neg p$
  - Let's try the truth table

p	q	$p \rightarrow q$	$\neg q$	$\neg p$	$\neg q \rightarrow \neg p$
T	T	T	F	F	T
T	F	F	T	F	F
F	T	T	F	T	T
F	F	T	T	T	T

Bidirectional Statements 1

- Let p and q be propositions. The **biconditional statement** $\boldsymbol{p \leftrightarrow q}$ is the proposition "$\boldsymbol{p}$ **if and only if** $\boldsymbol{q}$", or short as "$\boldsymbol{p}$ **iff** $\boldsymbol{q}$".
- The biconditional statement $p \leftrightarrow q$ is true when p and q have the same truth values, and is false otherwise.
- Truth table for biconditional statement

$\boldsymbol{p}$	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	**F**	**T**

Bidirectional Statements 2

- Another way to understand
  - $p \leftrightarrow q$ has exactly the same truth value as $(p \rightarrow q) \wedge (q \rightarrow p)$.
  - p is necessary and sufficient for q
    - E.g. "today is the Fourth of July" is a necessary and sufficient condition for "today is Independence Day in the United States"
- Example: Let p be the statement "You can take the flight" and q be the statement "You buy a ticket." Express the biconditional $p \leftrightarrow q$ as an English sentence.
  - Solution:
    "You can take the flight is necessary and sufficient for you buy a ticket" – correct but confusing. Or better:
    "You can take the flight if and only if you buy a ticket."

Precedence of Logical Operators

Operator	Precedence	Explanation
$\neg$	1	Highest Precedence
$\wedge$	2	
$\oplus$	3	
$\vee$	4	
$\rightarrow$	5	
$\leftrightarrow$	6	Lowest Precedence

- This is related to how to parse the expression
- You may **need parentheses**:
  - E.g. $(q \vee \mathrm{r}) \wedge (p \vee r)$

Truth Tables For Compound Propositions

- Example: Construct the truth table of the compound proposition $p \vee q \rightarrow \neg r$

p	q	r	$\neg r$	$p \vee q$	$p \vee q \rightarrow \neg r$
T	T	T	F	T	F
T	T	F	T	T	T
T	F	T	F	T	F
T	F	F	T	T	T
F	T	T	F	T	F
F	T	F	T	T	T
F	F	T	F	F	T
F	F	F	T	F	T

Applications of Propositional Logic 1

- Translating English Sentences
  - Identify atomic propositions and represent using propositional variables.
  - Determine appropriate logical connectives
- Example: "If I go to Harry's or to the concert, I will not go shopping."
  - p: I go to Harry's
  - q: I go to the concert.
  - r: I will go shopping.
  - If *p* or *q* then not *r*.
  - $(p \vee q) \rightarrow \neg r$

Applications of Propositional Logic 2

- Example: "You can access the Internet from campus **only if** you are a computer science major or you are not a freshman."
  - a: You can access the internet from campus
  - c: You are a computer science major
  - f: You are a freshman
  - Only if: necessary condition, so right side is the conclusion
  - $a \rightarrow (c \vee \neg f)$
- Example: "The automated reply cannot be sent **when** the file system is full"
  - p: The automated reply can be sent
  - q: The file system is full
  - $q \rightarrow \neg p$

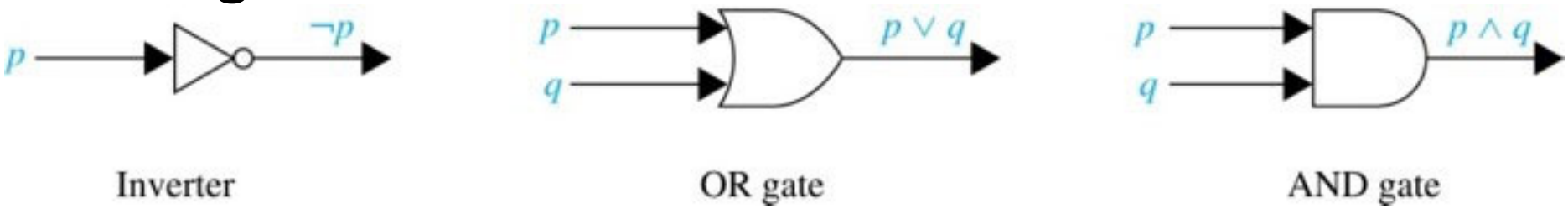
Can be used as system specifications

Applications of Propositional Logic 3

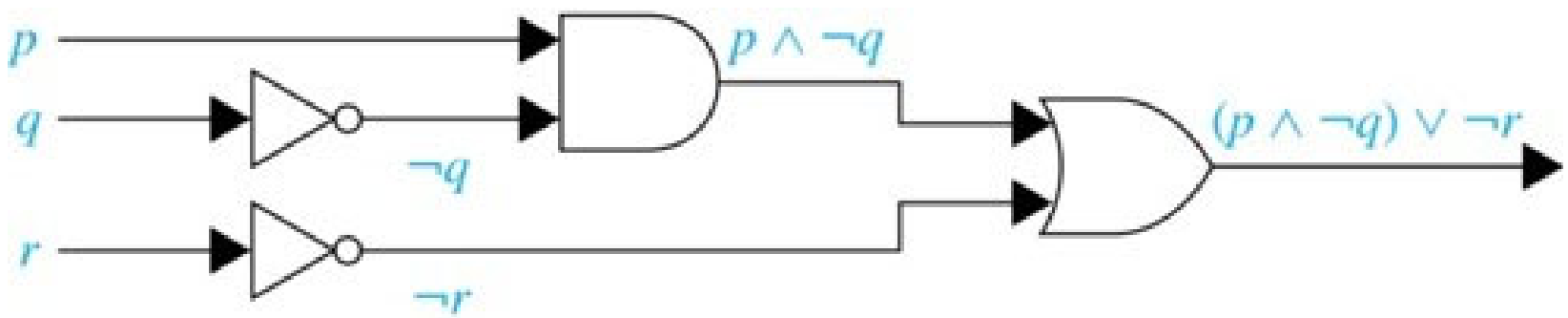
- Consistent System Specifications
  - A list of propositions is ***consistent*** if it is possible to assign truth values to the proposition variables so that each proposition is true.
  - A compound proposition is ***satisfiable*** if there is an assignment of truth values to its variables that makes it true.
- Example: Are these specifications consistent?
  - "The diagnostic message is stored in the buffer or it is retransmitted."
  - "The diagnostic message is not stored in the buffer."
  - "If the diagnostic message is stored in the buffer, then it is retransmitted."
    - p: The diagnostic message is stored in the buffer
    - q: The diagnostic message is retransmitted
    - We have: $p \vee q, \neg p, p \rightarrow q$
    - When p is false, all statements are true: **consistent**.
    - What if we add "The diagnostic message is not retransmitted"

Applications of Propositional Logic 4

- **Logic Circuits**



- A combinational circuit



Practices 1

- 1. Determine whether each of these conditional statements is true or false.
  - If 1+1=3, then unicorns exist.
  - If 1+1=3, then dogs can fly.
  - If 1+1=2, then dogs can fly.
  - If 2+2=4, then 1+2=3.

Practices 2

- 1. Determine whether each of these conditional statements is true or false.
  - If 1+1=3, then unicorns exist.
    - This is F→ F, which is true.
  - If 1+1=3, then dogs can fly.
    - This is F→F, which is true.
  - If 1+1=2, then dogs can fly.
    - This is T→ F, which is false.
  - If 2+2=4, then 1+2=3.
    - This is T→ T, which is true.

Practices 3

- 2. How many rows appear in a truth table for each of these compound propositions?
  - $(q \rightarrow \neg p) \vee (\neg p \rightarrow \neg q)$
  - $(p \rightarrow r) \vee (\neg s \rightarrow \neg t) \vee (\neg u \rightarrow v)$

Practices 4

- 2. How many rows appear in a truth table for each of these compound propositions?
  - A truth table will need 2^n rows if there are n variables.

  - $(q \rightarrow \neg p) \vee (\neg p \rightarrow \neg q)$
    - 2^2=4
  - $(p \rightarrow r) \vee (\neg s \rightarrow \neg t) \vee (\neg u \rightarrow v)$
    - 2^6=64

Practices 5

- 3. Write the truth table for the proposition $\neg(r \rightarrow \neg q) \vee (p \wedge \neg r)$

Practices 6

- 3. Write the truth table for the proposition $\neg(r \rightarrow \neg q) \vee (p \wedge \neg r)$

p	**q**	**r**	**$\neg(r \rightarrow \neg q) \vee (p \wedge \neg r)$**
T	T	T	T
T	T	F	T
T	F	T	F
T	F	F	T
F	T	T	T
F	T	F	F
F	F	T	F
F	F	F	F

Practices 7

- 4. Write each of these statements in the form of "if p, then q" in English.
  - It snows whenever the wind blows from the northeast.
  - It's necessary to walk eight miles to get to the top.
  - Your guarantee is good only if you bought your CD player less than 90 days ago.
  - Jan will go swimming unless the water is too cold.

Practices 8

- 4. Write each of these statements in the form of “if p, then q” in English.
  - It snows whenever the wind blows from the northeast.
    - If the wind blows from the northeast, then it snows.
  - It’s necessary to walk eight miles to get to the top.
    - If you get to the top, then you must have walked 8 miles.
    - (Necessary part is the then part.)
  - Your guarantee is good only if you bought your CD player less than 90 days ago.
    - If your guarantee is good, then you must have bought the CD player less than 90 days ago.
    - (Only if part is the necessary part, meaning the then part.)
  - Jan will go swimming unless the water is too cold.
    - If the water is not too cold, then Jan will go swimming.
    - (Unless means if not, so the unless part should be the negated if part.)

Self Test

- Consider the propositional variables as Boolean variables in C++, and find the equivalent operators in C++ for each logic connective.
- Give a proposition that has following truth table:

p	$\neg q$	?
T	T	F
T	F	F
F	T	T
F	F	F

- How many rows (w/ or w/o true result) are there for a given logical expression?
  - How many truth tables are there involving variables p and q?

Self Test Discussion 1

- 1. Basics: not (~), and (&), or (|), and exclusive or (^) http://www.cplusplus.com/doc/boolean/
  - Pay special attention to the C++ operator on exclusive or: $\wedge\ is\ different\ from\ \wedge$ --- ^ is “shift+6” on keyboard
  - A *bit* is the contraction of *binary digit*, a zero (0) or a one (1).
  - A ***bit string*** is a sequence of zero or more bits. The ***length*** of the string is the number of bits in the string.

Self Test Discussion 2

- Example: Find the bitwise *OR*, bitwise *AND*, and bitwise *XOR*, of the bit strings 01 1011 0110 and 11 0001 1101.
  - Solution:

OR	**0**	**1**	**1**	**0**	**1**	**1**	**0**	**1**	**1**	**0**
(|)	**1**	**1**	**0**	**0**	**0**	**1**	**1**	**1**	**0**	**1**
	1	1	1	0	1	1	1	1	1	1
AND	**0**	**1**	**1**	**0**	**1**	**1**	**0**	**1**	**1**	**0**
(&)	**1**	**1**	**0**	**0**	**0**	**1**	**1**	**1**	**0**	**1**
	0	1	0	0	0	1	0	1	0	0
XOR	**0**	**1**	**1**	**0**	**1**	**1**	**0**	**1**	**1**	**0**
(^)	**1**	**1**	**0**	**0**	**0**	**1**	**1**	**1**	**0**	**1**
	1	0	1	0	1	0	1	0	1	1

Self Test Discussion 3

- 2.

p	$\neg q$	?
T	T	F
T	F	F
F	T	T
F	F	F

- There is only one true case – conjunction is needed
  - $\neg p \wedge \neg q$

- 3. See the solutions in the textbook on page S-1 in the end

Extension

- Generate the truth table in a smart way
  - Traditional solution uses nested loops
    - Hard to maintain: no adaptivity
  - New solution
    - Extract the variables' values from the case number
      - N variables means 2^N cases: 0, 1, 2, 3, …, 2^N -1
      - For a case number, convert to its binary representation
        - E.g., when we have 4 variables (p, q, m, n) in the given compound expression, so there will be 16 cases. Let's use11 for illustration: 11 is $(1011)_2$ in binary, we can extract each bit from this binary representation – 1 means true, and 0 means false, so that we can have p = true, q = false, m = true, n = true